

Corporate Finance (in lingua inglese) - A. Y. 2025/2026 - Topic 5b: NPV and other investment criteria - solution

TEXT and SOLUTION (through the use of Excel)

The purchase price of a new machine is € 1,500. The additional revenues will be € 800 a year, the predicted incremental expenses are € 40 per year (not including depreciation) from year 1 until the end of the project. The machine will be depreciated over a period of 5 years, by € 250 per year. The firm expects an annual increase in inventories, receivables and payables for the project of, respectively, 10%, 5% and 4% of the annual additional revenues from year 1 to 2. The working capital will be zero in the fifth year. The sale price of the new machine at the end of the fifth year will be € 250. The cost of capital for this project is 10%, the tax rate is 35% and the project life is 5 years. Calculate the NPV, IRR and payback period of the project.

		Period					
Year		0	1	2	3	4	5
Panel A Capital Investment							
1	Cash flow from investment in fixed assets	-1500,00					
2	Sales of fixed assets						250,00
3	Less tax on sales						0,00
4	Cash flow from capital investment (1+2+3)	-1500,00					250,00
Panel B Operating Cash Flow							
5	Revenues		800,00	800,00	800,00	800,00	800,00
6	Expenses*		40,00	40,00	40,00	40,00	40,00
7	Depreciation		250,00	250,00	250,00	250,00	250,00
8	Pretax profit (5 - 6 - 7)		510,00	510,00	510,00	510,00	510,00
9	Tax at 35%		178,50	178,50	178,50	178,50	178,50
10	Profit after tax (8 - 9)		331,50	331,50	331,50	331,50	331,50
11	Operating cash flow (10 + 7)		581,50	581,50	581,50	581,50	581,50
Panel C Investment in Working Capital							
12	Working capital		88,00	176,00	176,00	176,00	0,00
13	Change in working capital		88,00	88,00	0,00	0,00	-176,00
14	Cash flow from investment in w. c . (-13)		-88,00	-88,00	0,00	0,00	176,00
Panel D Project Valuation							
15	Total project cash flow (4 + 11 + 14)	- 1.500,00	493,50	493,50	581,50	581,50	1.007,50
16	Discount factor at 10%	1,00	0,91	0,83	0,75	0,68	0,62
17	Discounted cash flows (15 x 16)	- 1.500,00	448,64	407,85	436,89	397,17	625,58
18	NPV (sum of discounted cash flows)	816,13					

* Not including depreciation

Alternative method for calculating the discounted cash flows (by putting all the data in a single formula)

Calculations of the discounted cash flows at 10%						
Period	0	1	2	3	4	5
Formula	$-1500,00/(1+10\%)^0$	$493,50/(1+10\%)^1$	$493,50/(1+10\%)^2$	$581,50/(1+10\%)^3$	$581,50/(1+10\%)^4$	$1007,50/(1+10\%)^5$
Discounted cash flows	-1500,00	448,64	407,85	436,89	397,17	625,58

For example, the present value at time 3 is calculated as follows: $= 581,50/POTENZA(1+10\%;3) = 436,89$

IRR

IRR = 27,2063% = TIR.COST(C34:H34), where, in this case, C34:H34 is the series of the total project cash flows.

If the IRR is correct, then the NPV at 27,2063% will be equal to 0:

Period	0	1	2	3	4	5
Formula	$-1500/(1+27,2063\%)^0$	$493,50/(1+27,2063\%)^1$	$493,50/(1+27,2063\%)^2$	$581,50/(1+27,2063\%)^3$	$581,50/(1+27,2063\%)^4$	$1007,50/(1+27,2063\%)^5$
Discounted cash flows	-1.500	388	305	283	222	302
NPV (sum of discounted cash flows)	0,00					

Payback period

Notice that $513 = 1500 - 493,5 - 493,5$ is the investment which is still to be recovered over year 3. $513/581,5$ is equal to about 0.8822. Therefore: the **PP** is about 2.8822 years, that is 2 years and $0.8822 \cdot 365$ days, which is equal to 2 years and about 322 days.

If the cut-off period is 3 years or more you will accept the project, whereas if it is 2 years or less you will refuse it.

Remember that the functions being used (POTENZA and TIR.COST) concern the Italian version of the spreadsheet (Excel). The English versions would be, respectively: POWER instead of POTENZA and IRR instead of TIR.COST.